

Installer Audit Rule Set

Installs business unit · v1 DRAFT · 2026-07-08 · for Ilias / Tim · internal

For Ilias / Tim. Drafted 2026-07-08. This is the "Installs" business-unit ruleset for the same ServiceTitan audit agent that already runs HVAC Sales, HVAC Service, Plumbing Sales, and Plumbing Service (the "Marketing Operating System" weekly + per-job Slack alerts). It plugs in the SAME way HVAC Service does: per install appointment, attributed to the technician (the install crew lead), one clean flag per real problem. Grounded in the Installer's Handbook v4 (onboarding/install-handbook, CH 7-13), the install-milestone rules already drafted for the PM agent (pm-agent-RULES.md R18-R22), and Jane's confirmed installer process (2026-07-08). Nothing here is invented; every rule cites its source.

HOW IT SLOTS IN (read this first)

- HVAC Service today has ONE rule: `hvac_payment_missing_on_completed_job`. Installs is richer because an installer owns a documented close-out (job status, forms, photos, equipment) AND attendance/labor duties (arrival, meal break). So this is a small set, not one rule. Ship the [core] ones first, watch for noise, then enable the rest.
- Same output shape as the other units: one Slack line per fail, tech-attributed, plus the weekly summary row (BU / ruleset / rule / severity / open / resolved).
- Precision over volume. When a record is ambiguous, or a field is missing from the payload (not from the job), SKIP. Do not guess. A false flag is worse than a missed one.

CONFIG THE AGENT NEEDS (one blank to fill)

- `install_business_unit_id(s)` = SET THIS. The weekly summary lists 1812 HVAC Sales, 1810 HVAC Service, 64326403 Plumbing Sales, 64315277 Plumbing Service. There is no Installs BU in the current output — Ilias needs the install BU id(s) (HVAC install, and Plumbing install if separate) to scope this ruleset.

CONFIGURABLE VALUES (defaults, tune later)

```
first_day_collect_pct   = 50   (% of invoice total due by end of install day 1)
final_day_collect_pct   = 100  (cumulative % due by end of the final install day)
deposit_reminder_lead   = 1    (remind this many days before install day 1 if no deposit)
completion_photos_min   = 1    (at least one install photo attached)
arrival_grace_min       = 15    (arrived "late" if this many min after scheduled start)
meal_break_after_hours  = 5     (CA: a 30-min meal is due before the end of the 5th hour)
second_meal_after_hours = 10    (CA: a 2nd 30-min meal is due before the end of the 10th hour)
meal_break_min_minutes  = 30    (a valid meal break is at least this many minutes)
end_of_day_grace        = same  install day (never flag "money not in yet" same day)
```

THE REQUIRED FORMS + JOB STATUS (the spine — confirmed by Jane 2026-07-08)

Every install job must end up CLOSED OUT: the job in status Completed, with two required forms both in status = completed. The audit is essentially "is each of these done at the point it should be." Each is gated to a moment in the job:

Homeowner Authorization Form — signed AFTER the crew arrives but BEFORE any installation work starts. The go / no-go gate on the job.

Installation Completion Form — filled at the END of the final install day.
Job status = Completed — set at the end of the final day. Installers forget this a lot,
which leaves finished jobs sitting open and unbillable.
If any of these is missing or left in a non-completed status past its gate, that is a fail.

THE RULES

Tag [core] = ship first. The rest enable after a dry-run confirms the field is on the ServiceTitan payload for install jobs. Rules are grouped: A close-out, B attendance/labor, C money, D documentation.

GROUP A: CLOSE-OUT (JOB STATUS + REQUIRED FORMS)

I1 `install_job_not_marked_complete` CORE HIGH

Fail when: the install work is finished but the ServiceTitan JOB STATUS is not Completed — i.e. the final install day / scheduled window has passed (or the Completion Form is done, or full payment is in) yet the job is still In Progress / Working / Dispatched / Scheduled. Installers routinely forget to close the job.
Skip when: a multi-day install is still within its scheduled span (not yet the final day). Only judge "should be complete" once the final day/window has passed.
Data: job status, scheduled final-day/window end, completion-form status, payment.
Source: Handbook CH10 (Project fully complete — YES / NO); Jane 2026-07-08.

I2 `install_completion_form_not_completed` CORE HIGH

Fail when: the final install day is done / the job is marked complete, and the Installation Completion Form is missing or not in status completed.
Data: install-complete flag (final day), completion-form status.
Source: Handbook CH10 (Installation Completion Form — end of final day); PM agent R20.

I3 `install_authorization_form_not_completed` CORE HIGH

Fail when: installation work has begun (crew arrived and the job is in-progress or complete) and the Homeowner Authorization Form is missing or not in status completed. The form must be signed after arrival and BEFORE work starts, so on any job that has started it should already be completed.
Data: crew arrival / job status (started vs not), authorization-form status.
Source: Handbook CH7 (authorization before starting the install); PM agent R19.

GROUP B: ATTENDANCE & LABOR (ARRIVAL + MEAL BREAK)

I4 `install_arrival_not_marked` CORE MEDIUM

Fail when: the appointment's scheduled start has passed (or the job is already in-progress / complete) and NO arrival / on-site timestamp is recorded for the crew. Installers forget to tap "arrived," which also breaks arrival-timing reporting.
Skip when: the appointment start is still in the future.
Data: arrival / on-site timestamp, scheduled start, job status.
Source: Handbook CH5-6 (arrival & approach); attendance standards CH1.

15 install_arrived_late MEDIUM

Fail when: the crew's arrival timestamp is more than arrival_grace_min after the scheduled appointment start.

Note: this is "after scheduled start," NOT the sales "first half of a 2-hour window" metric (installs are all-day, so that rule does not apply – see NOT AN INSTALLER RULE).

Skip when: arrival is not marked at all (that is I4, do not double-flag), OR the job was rescheduled and the new start is honored.

Data: arrival timestamp, scheduled start.

Source: Handbook CH1 (attendance & punctuality standards).

16 install_meal_break_not_taken CORE HIGH CA COMPLIANCE

California requirement, IRBIS is in San Jose. Fail when: a technician's install day runs longer than meal_break_after_hours (5) and no meal break of at least meal_break_min_minutes (30) is recorded for that tech that day; and a SECOND meal when the day runs longer than second_meal_after_hours (10). This is compliance, not just hygiene – treat as high.

Granularity: this is a PER-TECHNICIAN-PER-DAY check (from the timesheet / clock), not a per-job check. Attribute the flag to the technician + date.

Skip when: the day is under the threshold, OR a valid on-duty meal agreement applies (rare – confirm with HR whether IRBIS uses these).

Data: technician clock-in/out, recorded break events, hours worked that day.

Source: CA meal-period law; IRBIS San Jose (CLAUDE.md). CONFIRM with Jane/HR: where breaks are logged (ServiceTitan timesheet vs payroll) and whether the agent can read them.

GROUP C: MONEY (DEPOSIT + INSTALL-DAY COLLECTION)

17 install_deposit_not_collected_before_day1 CORE HIGH REMINDER

Fires PROACTIVELY on an UPCOMING install, before the crew starts – a reminder to collect the deposit, not an after-the-fact miss.

Fail / remind when: an install is scheduled to start (day 1 within deposit_reminder_lead days, or today) and NO deposit payment is recorded on the job yet.

Skip when: financed job (lender funds it, no customer deposit), OR the job is flagged deposit-waived / customer-arranged, OR the install date is outside the reminder window.

Data: deposit payment record + date, scheduled install start date, financing flag.

Source: Handbook CH11; PM agent R22 (Deposit before install).

IMPLEMENTATION NOTE: most rules look at completed / in-progress jobs. I7 looks FORWARD at scheduled installs, so the agent must also scan upcoming install appointments. Cleanest as a separate daily "installs starting tomorrow" pass.

18 install_payment_milestone_short CORE HIGH

The install version of hvac_payment_missing_on_completed_job, milestone-aware because installers collect in halves, not all at once:

- By END of install day 1: at least first_day_collect_pct (50%) of the invoice total.

- By END of the final install day: the full balance (final_day_collect_pct = 100%).

Fail (high) when: the final install day is over and the balance is still not fully paid.

Fail (medium) when: a multi-day install is past end of day 1 and less than 50% collected. (Enable this half after a dry-run confirms per-day appointment data.)

Skip when: financed job (lender covers the balance), OR still the same day as the milestone (end-of-day money), OR a single-day install still on its install day.

Note: a 50%-open balance mid-way through a multi-day install is EXPECTED, not a fail.

Data: invoice total, collected/balance, payment timestamps, financing flag, per-day appointments (day 1 vs final day).

Source: Handbook CH11 (half end of day 1, half end of final day); PM agent R18.

GROUP D: DOCUMENTATION (ENABLE AFTER FIELDS CONFIRMED)

19 install_photos_missing MEDIUM

Fail when: an install is marked complete with fewer than completion_photos_min installation pictures attached.

Data: attached-photo count, job status.

Source: Handbook CH10 (Pictures of installation).

110 install_materials_not_scanned MEDIUM

Fail when: an install is complete but no materials were scanned onto the job (the Golden

Rule: "make every physical material movement visible in Ply" – Ply syncs to the ST job).

Skip when: bare-labor / no-material job type.

Data: job material lines / Ply-sync count, job status.

Source: Install Tech Handbook (Ply) CH1 Golden Rule + CH8 material logistics.

111 install_equipment_not_registered MEDIUM

Fail when: an install is completed with equipment registration / equipment labels not done.

Data: equipment registration status, equipment-label field, job status.

Source: Handbook CH10 (equipment labels); PM agent R21.

112 install_review_not_requested LOW

Fail when: an install is completed and the review-requested step is empty.

Data: review-requested flag, job status.

Source: Handbook CH12 (Requesting client reviews). Low severity, enable last.

NOT AN INSTALLER RULE (important — do not copy these over)

- "Fewer than 3 options / Good-Better-Best" (sales_options_fewer_than_three): a SALES / diagnostic behavior. An installer executes an already-sold job; there is no options requirement. Do NOT run the 3-options rule on the Installs unit.
- "Arrived after first half of window" (sales_arrival_after_first_half): a sales-advisor metric on a 2-hour window. Installs are all-day / multi-day, so first-half timing is meaningless. The installer arrival check is I5 (arrived after scheduled start) instead.

EXCEPTIONS: DO NOT FLAG (carve-outs, same discipline as the Dispatcher agent)

- End-of-day money: on-site payment is collected at end of day. Never flag "payment not collected" earlier the same install day (applies to I8). Same as the HVAC Service carve-out.
- Financed projects: when the job is financed the lender pays the balance; a nonzero customer balance is NOT a missing-payment fail (applies to I7 and I8). Handbook CH11.
- Multi-day installs: a job open across several days is not "incomplete close-out" until the final day passes. Only judge I1/I2/I9/I11/I12 once the install should be done. And a 50%-open balance partway through is EXPECTED — do not flag it as a payment miss.
- Meal break under threshold: a short install day (under meal_break_after_hours) needs no meal break — do not flag it (I6).
- Customer-pulled permit / permit owned by PM: permit and HOA/asbestos live on the PM project record, not the installer's appointment — leave those to the PM agent, don't duplicate here.
- Out-of-scope jobs never evaluated: service call, warranty, recall, Home Care Plan maintenance, free diagnostic, internal/R&D.

SLACK OUTPUT (identical shape to the other units)

HIGH - Installs: Job Not Marked Complete

Technician: <crew lead>

Appointment: <date, window>

Arrived: <time>

Invoice: \$<total> total / \$<balance> balance

Issue: <one concrete sentence naming the actual value>

Action: <what to check / who to coach>

Open in ServiceTitan: <https://go.servicetitan.com/#/Job/Index/<id>>

The deposit reminder (I7) is FORWARD-LOOKING (fires the day before), so it reads as a heads-up rather than a violation:

REMINDER - Installs: Deposit Not Collected — install starts <date>, no deposit on file.

The meal-break flag (I6) is attributed to the technician + date, not a single job:

HIGH - Installs: Meal Break Not Recorded — <tech>, <date>: 9.5 hrs worked, no 30-min meal.

...and the weekly roll-up gets an "Installs" block:

BU ID: <install_business_unit_id> Ruleset: Installer Audit

- install_job_not_marked_complete [high] open: N
- install_completion_form_not_completed [high] open: N
- install_authorization_form_not_completed [high] open: N
- install_meal_break_not_taken [high] open: N
- install_deposit_not_collected_before_day1 [high] open: N
- install_payment_milestone_short [high] open: N
- install_arrival_not_marked [medium] open: N

...

SHIP ORDER (recommendation)

1. Turn on I1 / I2 / I3 first — job-complete status and the two required forms. These are the clearest, least-ambiguous fails and the spine of what Jane described.
2. Add I4 (arrival not marked) and I6 (meal break) next. I6 is a compliance win but needs its own data source (the timesheet, per-tech-per-day), so confirm the agent can read breaks before enabling. I5 (late) is lower priority — tune arrival_grace_min first.
3. Add I7 (deposit reminder) + I8 (payment milestone) together — both money, both need the deposit/financing fields; I7 also needs the forward scan of upcoming installs.
4. Then I9 / I10 / I11 once a dry-run confirms photos / material-scan / equipment fields are on the install payload. Hold I12 (review) for last.

OPEN QUESTIONS FOR TIM / ILIAS

- Install BU id(s) — single HVAC-install unit, or separate HVAC and Plumbing install?
- Form + status fields: does the job payload expose Homeowner Authorization Form status, Installation Completion Form status, and the job status transitions — or does that need a second endpoint (same gap that hid permits from the PM agent's first run)?
- Meal break data: where are technician breaks logged (ServiceTitan timesheet/labor vs the payroll system), and can the agent read clock-in/out + break events per tech per day?
- Per-day appointments: can the agent tell install day 1 from the final day, so I8's 50% end-of-day-1 half and the "final day passed" judgments can fire? If not, ship only the final-day full-balance half and key "should be complete" off the scheduled window end.
- Financing flag: which ServiceTitan field marks a job as financed, so I7 / I8 carve it out?
- Deposit vs the 50% (I7 + I8): is the deposit collected before day 1 SEPARATE from and on top of the 50%-end-of-day-1, or does it count toward that 50%? Needed so the two money rules don't double-count or contradict.
- "Some other things as well" (Jane, 2026-07-08): any remaining installer requirements to encode beyond the above (e.g. property-protection photos CH7, system-test / walkthrough sign-off CH11)?